University of Tehran

School of Electrical and Computer Engineering

Bachelor's Project Thesis

Electronics Engineering

Design and Implementation of a Real-Time Cardiac Arrhythmia Detection Device Using Machine Learning Algorithms

(Event Recorder)

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Dedicated to my dear and kind father, mother, and sister, who have always been compassionate, selfless companions and a firm, dependable source of support for me through life's hardships and difficulties.

Acknowledgments

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Abstract

Cardiovascular diseases are recognized as one of the leading causes of death worldwide, and timely diagnosis of cardiac rhythm abnormalities (arrhythmias) plays a fundamental role in preventing the serious complications of these diseases. Many arrhythmias occur intermittently and unpredictably and, in most cases, cannot be identified during the short period of electrocardiogram (ECG) recording or during echocardiography. In this context, continuous, long-term monitoring of the heart's electrical activity can provide an effective approach to the accurate and early detection of such abnormalities.

In this study, an intelligent wearable system was designed and implemented for continuous ECG monitoring of patients. The system comprises a front-end circuit for acquisition, filtering, and amplification of cardiac signals; a microcontroller equipped with an internal analog-to-digital converter; and an optimized machine-learning model implemented using TensorFlow Lite for instantaneous arrhythmia detection in 10-second ECG windows. When an arrhythmia is detected, the relevant portion of the signal, together with the temporal context before and after the event, is stored in the device memory to enable clinical evaluation by a physician.

Implementation results indicate that the system can provide continuous and reliable detection of arrhythmias, including premature ventricular contractions (PVCs), premature atrial contractions (PACs), and normal signals, while maintaining low power consumption. The proposed device may contribute to improving diagnostic accuracy, reducing unnecessary visits to healthcare centers, and enhancing the quality of care for patients with cardiac conditions.

Keywords: Cardiac arrhythmias, ECG, TensorFlow, cardiovascular diseases, analog-to-digital converter, machine learning, front-end circuit.

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Chapter One Introduction

1-1. Electrical Activity of the Heart

The human heart consists of four principal chambers: two atria and two ventricles. Precise coordination of contractions among these chambers is necessary for effective blood pumping. This coordination is controlled by an internal electrical conduction system. Unlike skeletal muscles, cardiac muscle (the myocardium) can automatically generate and conduct electrical potentials that ultimately produce muscular contractions.

Sinoatrial (SA) Node — The Heart's Natural Pacemaker

Electrical activity begins in the sinoatrial (SA) node, located in the upper wall of the right atrium near the entrance of the superior vena cava. This node contains specialized cells exhibiting automatic rhythmicity; in other words, they can generate regular action potentials without external stimulation.

The normal firing rate of the SA node is between 60 and 100 beats per minute, establishing the normal cardiac rhythm.

Conduction of the Signal to the AV Node

The action potential generated in the SA node spreads to the left atrium along interatrial pathways and travels through atrial tissue to the atrioventricular (AV) node. During this process, the atria contract first and pump blood into the ventricles.

The AV node produces a delay of approximately 0.1 seconds, allowing the atria to empty before the ventricles are stimulated. This delay is essential for effective cardiac pumping.

Transmission to the Ventricles: Bundle of His and Purkinje Fibers

After the AV node, the electrical potential travels through the bundle of His into the right and left bundle branches and then to the Purkinje fibers. This conduction network propagates the signal at a high speed (approximately 4 meters per second) throughout the ventricles and enables their coordinated contraction.

Action Potentials in Myocardial Cells

The action potential in cardiac muscle cells consists of five distinct phases:

- Phase 0 (depolarization): Rapid entry of sodium ions (Na^+) through voltage-gated sodium channels.
- Phase 1 (early repolarization): Sodium influx stops, accompanied by partial efflux of potassium ions (K^+).
- Phase 2 (plateau): Slow entry of calcium ions (Ca^{2+}) through L-type channels, balanced by potassium efflux.
- Phase 3 (final repolarization): Calcium channels close while potassium efflux continues.
- Phase 4 (resting potential): Return to the normal polarization state in preparation for the next action potential.

This electrical phenomenon appears in the ECG as the P, QRS, and T waves, each representing a part of the heart's electrical activity, as illustrated in Figure 1.

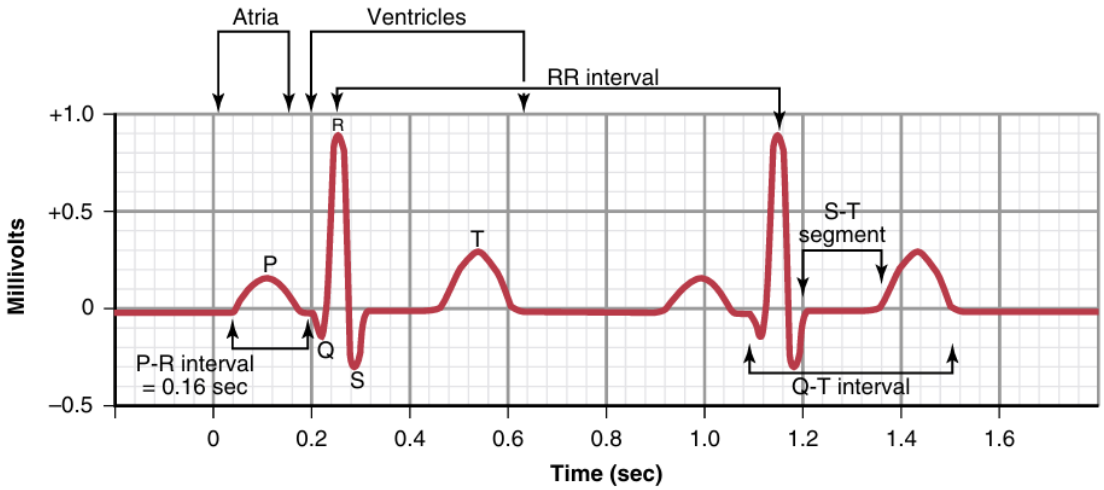


Figure 1. P, QRS, and T waves in an ECG signal.

1-2. Definition and Types of Cardiac Arrhythmias

Definition of Arrhythmia

An arrhythmia is any disturbance of the normal cardiac rhythm that may arise from abnormalities in the generation, conduction, or coordination of electrical signals. Arrhythmias may manifest as excessively fast beats (tachycardia), excessively slow beats (bradycardia), or irregular rhythms.

Based on their origin, arrhythmias are divided into two main categories:

- Atrial arrhythmias: originating in the atria or the SA/AV nodes.
- Ventricular arrhythmias: originating in the ventricles, bundle branches, or Purkinje fibers.

Some common arrhythmias are described below.

Premature Ventricular Contraction (PVC)

A premature ventricular contraction (PVC) is one of the most common ventricular arrhythmias and occurs as an early contraction of one or more ventricular regions. It originates in the Purkinje fibers or ventricular myocardium.

In an ECG, PVCs are characterized by wide, abnormal QRS complexes, generally without a preceding P wave. A normal heartbeat often resumes after a compensatory pause.

Possible causes include stress, caffeine, alcohol, ischemic heart disease or cardiomyopathy, and electrolyte disturbances, particularly potassium or magnesium deficiency.

Although a PVC may be benign in healthy individuals, frequent PVCs or their occurrence together with other arrhythmias may be a warning sign requiring further evaluation. Figure 2 presents an example of an ECG containing a PVC.

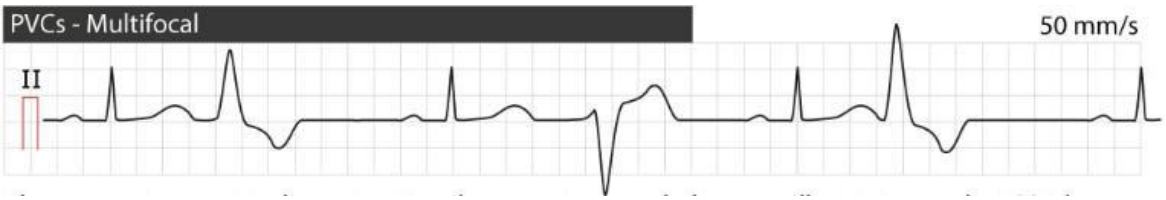


Figure 2. ECG signal exhibiting a premature ventricular contraction (PVC).

Premature Atrial Contraction (PAC)

A premature atrial contraction (PAC) is another type of arrhythmia that originates in the atrium outside the SA node. It results from premature stimulation at an ectopic focus within atrial tissue.

In the ECG, an early P wave with a morphology different from that of the normal P wave can be observed. The ensuing ventricular rhythm may remain normal or may include a short pause.

Possible causes include anxiety, nervous stimulation, consumption of coffee or nicotine, atrial diseases, and hormonal disturbances.

PACs may also occur in healthy individuals and are usually benign. In certain cases, however, they may precede more serious arrhythmias, such as atrial fibrillation (AF).

Figure 3 shows an example of an ECG signal containing a PAC.

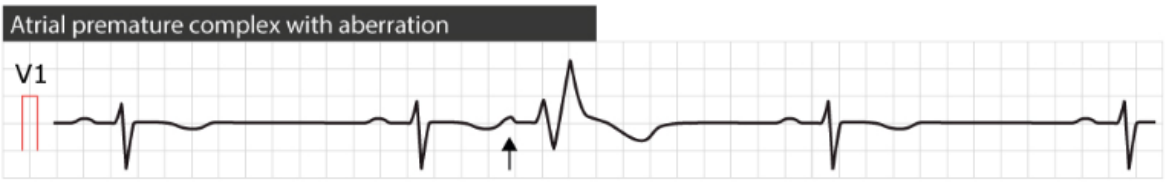


Figure 3. ECG signal exhibiting a premature atrial contraction (PAC).

The characteristics and comparison of different arrhythmias are summarized in Table 1.

Table 1. Comparison of different cardiac arrhythmias.

Arrhythmia type	Origin	ECG characteristics	Consequences
Sinus tachycardia	SA node	Heart rate > 100 bpm	Usually physiological
Bradycardia	SA node	HR < 60; prolonged PR and RR intervals	May be dangerous
PVC	Ventricle	Wide QRS without preceding P wave	Sometimes benign or a warning sign
PAC	Atrium	Premature P wave; normal QRS	Usually harmless
Atrial fibrillation (AF)	Atria	Irregular rhythm; no distinct P wave	Stroke; blood clots
Ventricular tachycardia (VT)	Ventricle	Repeated wide QRS complexes; no P wave	Life-threatening

1-3. Motivation for the Research and Device Design Objectives

Clinical Need for Continuous Cardiac-Rhythm Monitoring

In clinical examinations, physicians use methods such as electrocardiography (ECG) and echocardiography to evaluate a patient's cardiac condition. These tools provide useful information about the electrical and structural activity of the heart. However, a principal limitation is that the recordings are obtained over a very short period, usually a few seconds. Many cardiac rhythm abnormalities, referred to as arrhythmias, occur temporarily and intermittently and may not be present during an ECG recording or echocardiographic examination.

Patients experiencing syncope (temporary loss of consciousness), sudden palpitations, or chest discomfort may obtain normal results during an initial examination. However, such findings do not necessarily indicate that they are completely healthy.

Continuous, long-term ECG monitoring is needed so that any arrhythmic event can be recorded, analyzed, and evaluated when it occurs.

Existing Gap and the Idea Behind the Device

In response to this clinical need, the present project focuses on the design and construction of an intelligent wearable device capable of acquiring, processing, and analyzing cardiac signals around the clock. The device is based on a multi-part electronic system comprising:

- A front-end module to acquire, filter, and amplify the ECG signal.
- The microcontroller's internal analog-to-digital converter (ADC) to digitize the signal.
- Execution of a TensorFlow Lite artificial-intelligence model on the microcontroller.
- Automatic identification of time windows suspected of containing arrhythmias and storage of those windows in memory.
- The ability for a physician to review recorded episodes, confirm the diagnosis, and make treatment decisions.

Advantages of the Designed System

In addition to addressing the challenges of continuous monitoring, the design offers the following main advantages:

- Early detection of arrhythmias through algorithmic signal analysis, even in the absence of visible symptoms.
- Reduced costs and healthcare visits through less frequent testing in hospitals or clinics.
- High detection accuracy through trained neural networks capable of identifying hidden patterns in ECG signals.
- Usability in different environments because of the device's light and portable construction.
- Retrospective review by a physician through the recording of arrhythmic episodes along with the time context before and after each event.

The ultimate objective of this research is to design and implement an intelligent, reliable system for long-term monitoring of patients suspected of cardiac arrhythmias. Such a system should be able to monitor cardiac activity continuously without hospitalization, identify dangerous abnormalities, and provide essential data to physicians for detailed analysis and final confirmation.

Chapter Two Literature Review

With the development of artificial-intelligence technologies, the use of deep-learning models to analyze vital signals, particularly ECG signals, has brought about a fundamental change in automatic cardiac-arrhythmia detection. This chapter examines two reputable scientific papers that employ different modern approaches to heartbeat classification. Finally, their findings are compared with the system developed in this project.

2-1. Review of the First Paper

Article Title: A Deep Convolutional Neural Network Model to Classify Heartbeats

This article examines ECG analysis using convolutional neural networks (CNNs). The researchers present a deep CNN-based model for classifying heartbeats into five different classes of arrhythmia.

Research Structure and Methodology

Dataset: The MIT-BIH Arrhythmia Database, containing ECG recordings of real patients with detailed medical annotations.

Preprocessing: The ECG signal is divided into heartbeat-related windows. Each sample is extracted from a specific interval that includes the portion before and after the R peak.

Network Architecture

The model includes several convolutional layers, batch-normalization layers, pooling layers, and finally dense layers.

CNN layers are used to extract local features and can effectively identify subtle patterns in ECG signals.

Classification: Five classes are considered: normal beats, PVCs, atrial beats, ventricular beats with particular morphologies, and fusion beats.

Results

The overall accuracy of the model is notably high, and for many classes it outperforms traditional methods and even conventional machine-learning algorithms.

The authors show that a CNN can extract features automatically, eliminating the need for manual feature engineering.

Strengths

- A deep architecture with strong generalization ability.
- Testing on real clinical data.
- Automatic feature extraction without manual intervention.

Limitations

- The model is large and has many parameters, making it unsuitable for real-time execution on low-power hardware.
- Learning is based only on spatial information in the ECG signal, without consideration of the temporal dynamics of heartbeats.

2-2. Review of the Second Paper

Article Title: ECG Heartbeat Classification: A Deep Transferable Representation

This article focuses on using transfer-learning models to classify heartbeats from ECG signals. The authors seek to design a model that not only achieves high arrhythmia-classification accuracy but also transfers well across different patients.

Model Structure and Approach

Dataset: The MIT-BIH database is again used, but with particular emphasis on training on one portion of the data and testing on new patients to assess generalization.

Model Architecture

The principal model uses a hybrid deep architecture containing convolutional layers to extract spatial features, followed by dense layers for classification.

The principal innovation is a layer called the Representation Layer, which serves as a feature vector transferable across patients.

The main objective is to obtain a vector representation of a heartbeat that can be reused for different patients, even when their cardiac conditions differ.

Innovations

Transferable features: Unlike many models that perform well only on data from a specific patient, this model is designed for use with new patients without retraining.

Reduced need for large datasets: By learning a generalizable representation, the model requires less extensive patient-specific data.

Results

- High classification accuracy across five arrhythmia classes.
- Better performance than traditional approaches such as SVM and Random Forest.
- Strong resistance to overfitting and ability to operate on previously unseen data.

Strengths

- Suitability for real-time and mobile-health implementations.
- A more compact and optimized structure than the model in the first paper.

- Intelligent design of intermediate layers for the transfer of information.

Weaknesses

- In its original form, the model still cannot run on very limited microcontrollers and requires a lighter version.
- Accuracy is slightly lower for some rare heartbeat classes.

2-3. Position of the Present Research Relative to Previous Studies: Contribution and Innovation

Based on the two papers reviewed, the development of artificial-intelligence approaches to arrhythmia detection can be analyzed as follows.

The first paper emphasizes automatic feature extraction by CNNs and high accuracy in heartbeat classification, but its hardware implementation and real-time execution remain limited.

By introducing transferable representations, the second paper takes an important step toward models suitable for operational systems and long-term monitoring.

The system developed in this thesis attempts to integrate accurate hardware design, signal processing, and lightweight artificial intelligence for arrhythmia detection. Its main innovations relative to previous research are described below.

Practical Implementation on Wearable Hardware

Unlike the computationally intensive models in the reviewed articles, this project uses an optimized TensorFlow Lite model deployed on a low-power microcontroller, enabling real-time analysis of ECG signals in 10-second windows.

Real-Time Arrhythmia Detection

The implemented model can analyze cardiac activity continuously and store the relevant signal segment when arrhythmias such as PVC and PAC are detected. This capability was not implemented on operational hardware in either of the reviewed papers.

Integrated Hardware/Software System Design

The project is not limited to software modeling; a precise front-end circuit has also been developed to acquire and preprocess the ECG signal. This section includes a filter, amplifier, and ADC, which form the basis for correct operation of the artificial-intelligence algorithm.

Targeted Data Storage for Clinical Review

Unlike models that produce only a classification output, this system saves the portions of the signal before, during, and after an arrhythmia so that the physician can examine the event and its temporal context in detail.

The literature review indicates that deep-neural-network approaches can be effective and reliable for detecting cardiac arrhythmias. Nevertheless, their implementation in actual wearable devices presents challenges, including power consumption, model size, and hardware limitations.

By combining careful hardware design with a lightweight machine-learning model, the present project seeks to narrow this gap and bring artificial-intelligence-based arrhythmia detection into practical, microcontroller-level, real-time operation.

Chapter Three Hardware Section

3-1. Overall Block Diagram

Figure 4 presents a general block diagram of the automatic cardiac event-recording system. The different components of this diagram are explained below.

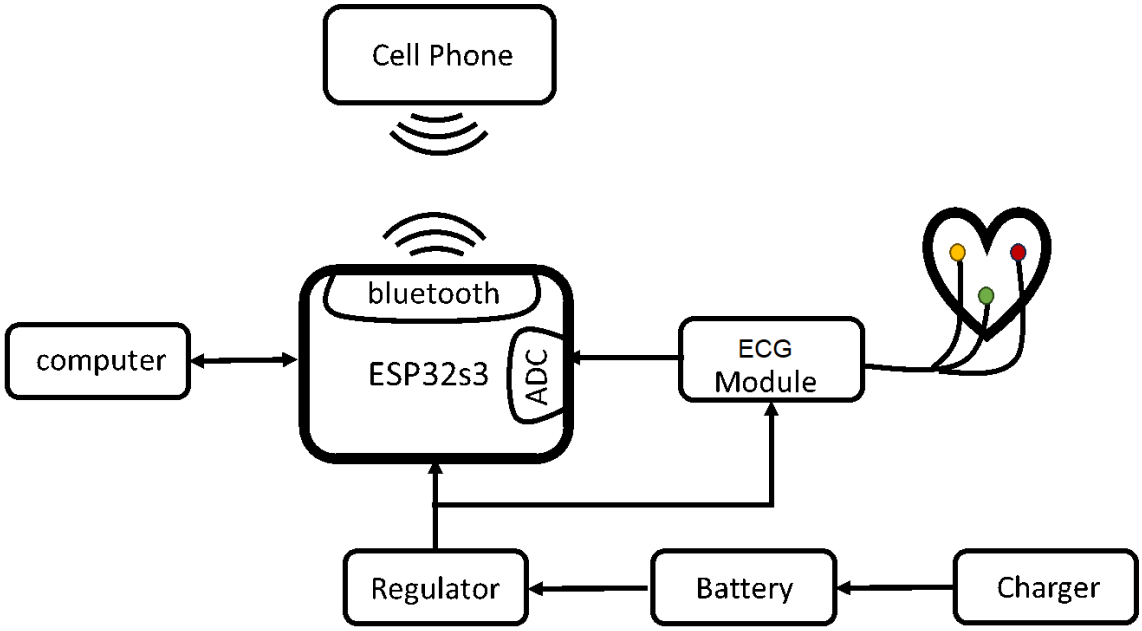


Figure 4. Block diagram of the automatic cardiac event recorder.

3-2. ESP32-S3 Microcontroller

The ESP32-S3 microcontroller, manufactured by Espressif in China, supports Bluetooth Low Energy (BLE). The artificial-intelligence arrhythmia-detection algorithm is implemented on this controller. It supervises and controls the different system sections, including acquisition of the amplified and filtered cardiac signal from the ECG module, Bluetooth communication when a connection to a mobile phone is needed, and connection and data exchange with a computer or laptop. Its advantages include the following.

1. The microcontroller contains a radio-frequency (RF) section that supports Wi-Fi and Bluetooth technologies. Mobile connectivity and data exchange are included in the device design, so the built-in Bluetooth support eliminates the need for a separate Bluetooth communication module.
2. Support for a high clock frequency of up to 240 MHz enables fast data processing and short processing times.
3. The microcontroller offers substantial memory capacity (520 kB of internal SRAM, 8 MB of external SRAM, and 16 MB of external flash), which is a significant advantage among available microcontrollers.
4. It supports an Ultra Low Power mode, which substantially reduces current drawn from the battery when the device is not processing data.
5. Its price is lower than that of microcontrollers with comparable characteristics, reducing the final cost of the device.

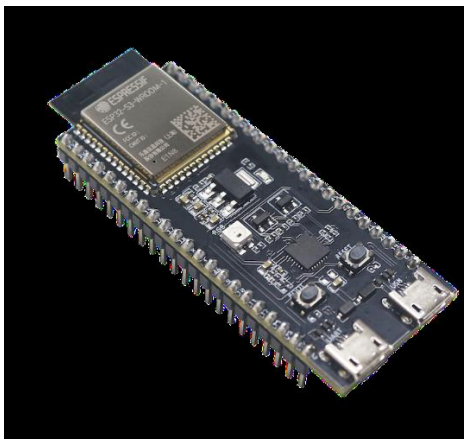


Figure 5. ESP32-S3 microcontroller development board.

3-3. ECG Module

Accurate and reliable extraction of the heart's electrical signal is one of the first and most essential steps in the design of this system. An ECG module is used for this purpose. The module is specifically designed for measuring, amplifying, and filtering ECG signals in low-power applications.

The ECG module connects to three surface electrodes on the patient's body (one positive, one negative, and one reference/ground electrode), shown in Figure 6. These electrodes pick up the very weak electrical signals produced by cardiac activity. The acquired signal first enters an internal precision instrumentation amplifier, which increases its amplitude to a suitable level for processing. This differential amplifier has high input impedance and low input noise, making it appropriate for biological signals such as ECG.

After amplification, the signal passes through an analog filtering stage. High-pass and low-pass filters with selected cutoff frequencies remove a large proportion of environmental interference, particularly motion artifacts and high-frequency noise. One of the module's key advantages is its internal Right Leg Drive (RLD) circuit, which uses active feedback to substantially reduce common-mode noise, especially interference from the 50 Hz mains supply. Such noise is one of the principal challenges in acquiring ECG signals.

The final output of the ECG module is an amplified, filtered signal ready for sampling. It is connected directly to the microcontroller's analog-to-digital converter (ADC) for subsequent digital processing. Given its very low power consumption, compact dimensions, and reliable performance in noisy environments, the module is an ideal choice for a wearable cardiac-monitoring device.



Figure 6. Electrodes used to acquire the cardiac signal.

3-4. TPS73233DBVR Voltage Regulator IC

The TPS73233DBVR is a precision, very-low-noise linear voltage regulator manufactured by Texas Instruments and widely used in sensitive analog and medical designs, including ECG monitoring systems.

In this cardiac-monitoring device, the TPS73233DBVR is used to stabilize the 3.3 V supply. Its function is to provide a highly accurate, stable, low-noise output for circuit sections sensitive to power-supply interference, particularly the analog ECG front end and the microcontroller's ADC.

This regulator is a Low Dropout (LDO) device, able to operate stably with a very small difference between input and output voltages (for example, a 3.5 V input and 3.3 V output). Another important characteristic is its very low output noise, approximately 30 µV RMS, and very small ripple. These properties make the device particularly suitable for medical applications in which electrical signal quality is essential.

The TPS73233 also has an Enable pin that permits software control of the regulator's on/off state. This feature is very useful in low-power wearable systems such as the wearable ECG device because it helps reduce overall system power consumption.

In summary, the TPS73233DBVR was selected for its combination of voltage stability, low noise, low power consumption, and small dimensions. All of these features are essential to the power-supply design of the ECG system's sensitive analog section.

3-5. Battery

Providing stable, low-noise energy to the circuit is a key factor in reliable continuous ECG monitoring, particularly because the device is wearable and must operate for long periods. A rechargeable lithium-ion battery with a nominal voltage of 3.7 V and a capacity of 1,200 mAh is used as the main power source. Its high energy density, low weight, rechargeability, and appropriate size make it an attractive choice for portable and wearable medical equipment.

The main design objective was to allow the device to operate for up to seven days without recharging. To achieve this, power consumption was considered throughout the circuit, especially along the supply path. The use of low-power microcontrollers supporting light-sleep and deep-sleep modes, efficient LDO regulators, and low-current components forms part of these considerations.

The 3.7 V battery output is regulated by the TPS73233DBVR to provide a stable, low-noise 3.3 V supply to sensitive circuit blocks, including the ADC, analog amplifiers, and microcontroller. This arrangement is intended to preserve supply quality and stability as the battery voltage gradually decreases.

Finally, appropriate battery selection and optimization of the supply path play an important role in meeting the project's principal objective: continuous and reliable ECG monitoring for several days without user intervention.

3-6. STM6600DQ25DM6F IC

Accurate power control and intelligent switching of circuit sections are particularly important in battery-powered wearable systems. For this purpose, the designed device uses the STM6600DQ25DM6F from STMicroelectronics. This component is an ultra-low-power on/off controller designed specifically for devices activated by a momentary push button.

In this project, the STM6600 provides intelligent power management for the main device circuitry. With a small button, it enables safe, controlled switching of battery-powered components without requiring a permanently actuated mechanical switch. This improves safety and battery life while simplifying the mechanical design.

One important feature of the STM6600 is its configurable delay in switching the system on and off. When the power button is pressed, the IC activates its output after a short delay or debounce interval and directs power to the supply circuit. For shutdown, the button must be held for a specified period, preventing accidental power-off caused by an unintended touch.

Because the ECG monitor is intended to operate for long intervals, for example seven days, careful power management is necessary. The STM6600 has standby current in the nanoampere range, so the main circuit can be activated only when needed.

At other times, the device can remain off or in a low-power state. This is consistent with the system's low-power, real-time architecture and contributes to its operating-time objectives.

Overall, the STM6600DQ25DM6F acts as the core of the device's power-management function, receiving the button input and controlling the power supply. It provides a reliable, safe, and optimized method for turning the wearable ECG monitor on and off while maintaining simple operation, extending battery operating time, and improving system stability.

3-7. FT232RL IC

In embedded systems that exchange data with a computer, a stable, fast, and simple communication interface between the host computer and the device microcontroller is essential. This project uses the FT232RL from FTDI for that purpose. The FT232RL is a USB-to-UART (serial) converter that translates digital USB 2.0 communication to the standard UART serial protocol.

Its main function is to connect the wearable ECG device to the physician's or user's personal computer. When the device is connected using a USB cable, the FT232RL is recognized by the operating system as a virtual COM port. This permits the user or physician to access data stored on the device using simple applications such as a serial terminal, without requiring specialized software.

The FT232RL supports data rates of up to 3 Mbaud and provides suitable transfer speed and reliability for ECG files or data packets. It also includes features such as a regulated 3.3 V output capable of supplying up to 50 mA and control signals including DTR, RTS, and CTS, making it appropriate for general-purpose serial communication.

In the present project, the FT232RL is used to read data recorded in the device's flash memory after arrhythmia detection and for the initial programming of the device. It therefore provides a standard connection for transferring recorded ECG data from the device to a computer for more detailed analysis by a physician.

Because the wearable ECG device operates from a battery and is generally not connected to a computer, USB communication is used only when medical data need to be reviewed or the device firmware must be updated. In these circumstances, the FT232RL can also obtain power for certain circuit sections through the USB port, simplifying the circuit design.

Overall, the FT232RL serves as a communication bridge between the external world (the physician's computer) and the device's internal microcontroller. Its high reliability, conformity with USB standards, and broad compatibility with operating systems make it a practical choice for serial communication in portable medical applications.

3-8. LTC4054LES5 IC

A reliable, compact, low-power charging circuit is a key requirement in wearable devices powered by lithium-ion batteries. In this project, the LTC4054LES5 is used to charge a 3.7 V, 1,200 mAh lithium-ion battery. It is a stand-alone, single-cell linear charger with current control, automatic charge termination, and protection against overcharging.

The LTC4054 is fully integrated and requires few external passive components, such as a resistor that sets the charging current. Its charging algorithm operates in two stages: an initial constant-current stage followed by a constant-voltage stage as the battery voltage approaches 4.2 V, helping prevent battery damage.

The LTC4054 is powered through the USB connector or an external supply. When the device is connected to a computer USB port or a charging adapter, the IC starts charging automatically and indicates its status using the CHRG pin (an open-drain output). This pin is connected to an LED that visually informs the user of the battery charging state (charging, completed, or fault).

An important advantage of the LTC4054LES5 is that it manages charging autonomously without requiring a microcontroller or external control system. This is particularly valuable for a wearable ECG monitor because charging can take place independently of the central processing unit.

The charging current is adjusted through the resistor programmed at the IC's PROG pin so that the battery charges in an appropriate period without excessive heating or accelerated chemical degradation. This contributes to improved battery life and overall safety.

In summary, using the LTC4054LES5 provides a reliable and efficient battery-charging solution that is compatible with the goals of a compact, low-power wearable ECG device and does not require a heat sink.

3-9. Push Buttons

A simple, effective, and reliable physical user interface is important for direct user interaction with the continuous ECG monitor. Three push buttons with independent functions are incorporated into the device hardware. Each performs a specific task and was selected and designed to occupy little space while maintaining precise, responsive, reliable operation.

The first button is the main Power Button and is connected directly to the STM6600DQ25DM6F. That IC controls the on/off sequence. The button operates momentarily: when it is held for the specified duration, the IC enables or disables the supply to the main circuit. This arrangement improves safety, reduces standby power consumption, and prevents unintended on/off operation.

The second button is the Reset Button, connected directly to the microcontroller's reset pin. In the event of an error or possible firmware lockup, the user or physician can restart the system without physically disconnecting its power supply. This function is particularly important for maintaining reliable operation in clinical and home environments.

The third button activates Bluetooth. When pressed, it signals the microcontroller to enable the Bluetooth communication function or enter discoverable mode. This allows data transmission to a smartphone or physician monitoring system only when required. It improves data security and reduces power consumption by avoiding continuous Bluetooth operation and associated interference with other devices.

Hardware and software debouncing techniques were used in the button design. Pull-up resistors and smoothing capacitors were added in hardware, while timing delays in software prevent a single press from being registered repeatedly. The physical positions of the buttons on the PCB and device enclosure were also chosen to permit easy use and reduce accidental activation.

In summary, the design of these three buttons plays an essential role in energy management, operational control, and communication, contributing to the simplicity, safety, and reliability of the wearable ECG device's physical interface.

3-11. System Printed Circuit Board

Figure 7 shows the PCB layout of the automatic cardiac event recorder designed using Altium Designer.

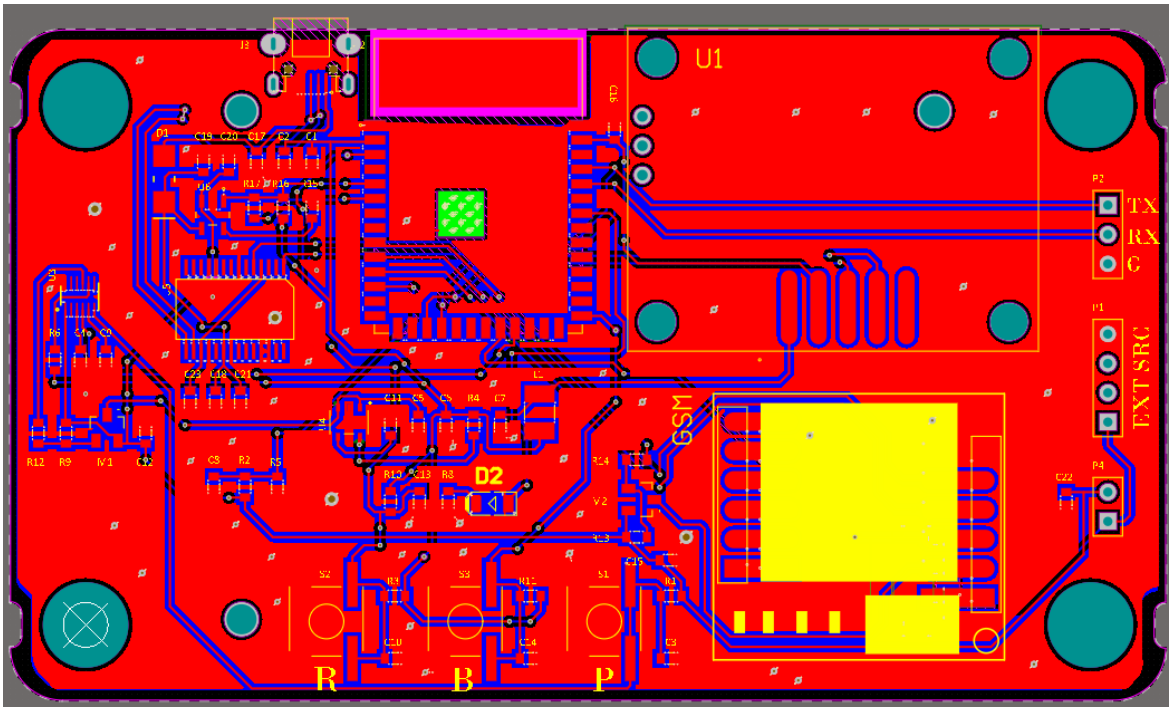


Figure 7. Printed circuit board of the system.

3-12. Fabricated and Assembled Development Board

The fabricated and assembled development board is shown in Figure 8.

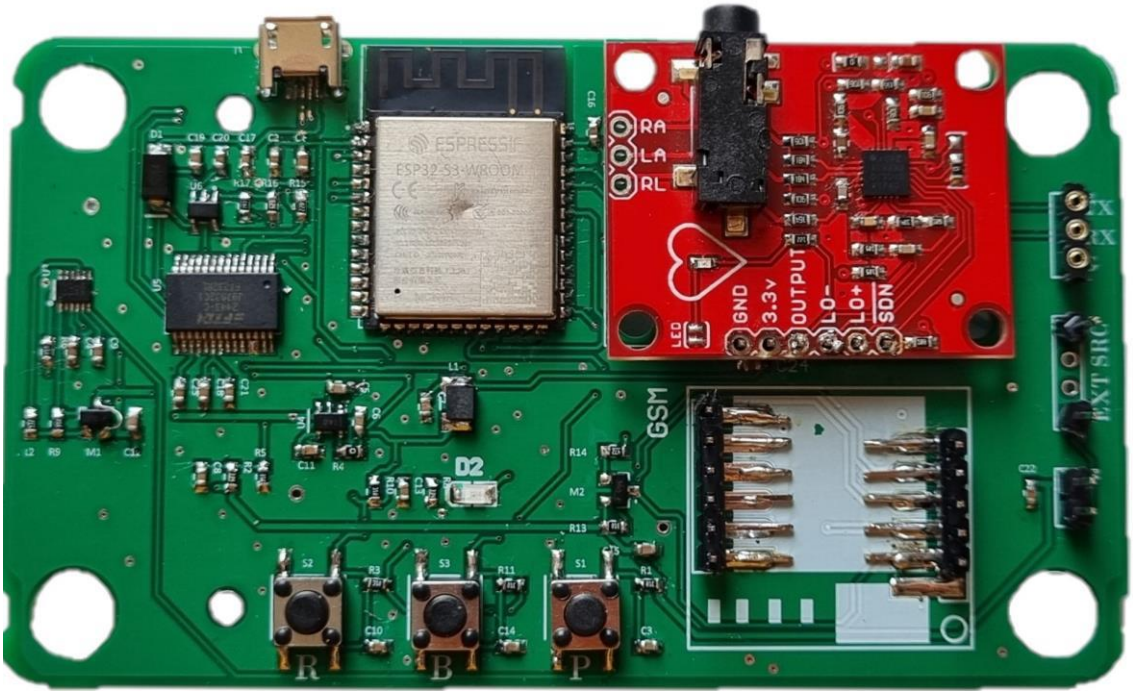


Figure 8. Assembled development board.

Chapter Four Software Section

The software of this system consists of several sections. Each section is examined separately and in detail below. The first is data acquisition.

4-1. Data Acquisition

The first stage of the process is the acquisition of data using the microcontroller's analog-to-digital converter (ADC).

Data acquisition is one of the most important parts of electronic and medical systems and directly affects their accuracy and efficiency. During this process, analog signals (such as a patient's cardiac signal) are collected using specialized modules and then converted into digital values by the ESP32-S3 microcontroller's ADC. These data are subsequently prepared for processing, analysis, and storage.

The specifications of the microcontroller ADC are presented in Table 2.

Table 2. ESP32-S3 microcontroller analog-to-digital converter (ADC) specifications.

Parameter	Value
Input range	0–3.3 V
Resolution	12 bits
Accuracy	0.8 mV

The ECG signal at the output of the ECG module is sampled by the ESP32-S3 microcontroller's ADC at a frequency of 195 Hz.

This process is performed using the RTC unit and the Ultra Low Power (ULP) coprocessor, which are introduced in the following sections.

4-1-1. Role of the RTC Unit in Data Acquisition

The RTC unit in the ESP32-S3 is a dedicated subsystem consisting of the following parts.

RTC Memory

This is a dedicated region of RAM that remains active in low-power modes, including Deep Sleep and Light Sleep, and preserves essential information.

RTC Timer

The timer can operate in low-power modes and can schedule events such as waking the main processor or the ULP coprocessor.

Internal Controllers

These include voltage management, low-power mode management, and other functions that participate in the ESP32-S3's different operating states.

Role of the RTC in Data Acquisition

Low-power mode management: The RTC controls low-power states, such as Light Sleep and Deep Sleep. In these states the main CPU is turned off, whereas the RTC remains active and can coordinate essential processes, including periodic ADC sampling.

ULP activation: The RTC can enable the ULP coprocessor and schedule it to perform simple tasks, such as reading ADC values.

Data storage: Data acquired by the ULP can be saved in RTC memory for subsequent processing when the main CPU wakes up.

4-1-2. Role of the ULP Coprocessor

The ULP is a separate low-power processor in the ESP32-S3 that operates independently of the main CPU. It remains active during low-power modes and can perform simple tasks. In the data-acquisition process, its role is as follows.

Periodic Sampling

The ULP can periodically read data from the ADC and store them in RTC memory. This can be done without waking the main CPU.

Reduced Power Consumption

Using the ULP allows the main CPU to remain in Light Sleep and become active only when the data are ready for processing. This significantly reduces energy consumption, a particularly important consideration in this battery-powered project.

RTC Scheduling

The RTC can schedule the ULP to perform specific functions, in this case ADC sampling. For example, it can activate the ULP at defined intervals to acquire ADC data, allowing the sampling frequency to be controlled.

4-1-3. Cooperation Between the RTC and ULP in Data Acquisition

The cooperation between the two units proceeds as follows.

Setting the Low-Power Mode

The RTC activates Light Sleep. In this mode, the main CPU is inactive while the RTC and ULP remain active.

Scheduling the ULP Through the RTC

The RTC schedules the ULP to perform specified tasks. Here, the ULP reads ADC data once every 5.12 milliseconds (approximately 195 Hz) and stores the readings in RTC memory.

Sampling by the ULP

The ULP reads the analog data through the ADC and stores the corresponding digital values in RTC memory.

Waking the Main CPU

After a specified period, here 10 seconds as scheduled by the RTC, the main CPU wakes, reads the data stored in RTC memory, and processes them.

4-1-4. Advantages of Using the RTC and ULP

Reduced Power Consumption

The main CPU is activated only when required, significantly reducing energy use. In this design, during each 10-second interval, the CPU is in sleep mode for approximately 9.5 seconds and active for about 0.5 seconds for processing and data storage. This greatly reduces average battery current and average power consumption.

Accurate Timing Management

The RTC can provide accurate scheduling for ULP activities and ensure that data are collected periodically and without interruption.

Data Retention in Low-Power Mode

Data acquired by the ULP are stored in RTC memory so that they are not lost while the main CPU is in a low-power mode.

Extended Battery Life

Using the RTC and ULP reduces energy consumption, an especially important advantage for portable medical devices.

In summary, the ESP32-S3 RTC acts as a system manager that controls low-power modes and schedules the ULP coprocessor to sample the ADC at 195 Hz in 10-second intervals. Their cooperation allows the system to acquire data periodically while consuming very little power. These features make the arrangement well suited to portable medical devices.

An example of a normal 80 bpm ECG signal generated by a simulator and acquired through the ADC at 195 Hz over a 10-second interval is presented in Figure 9.

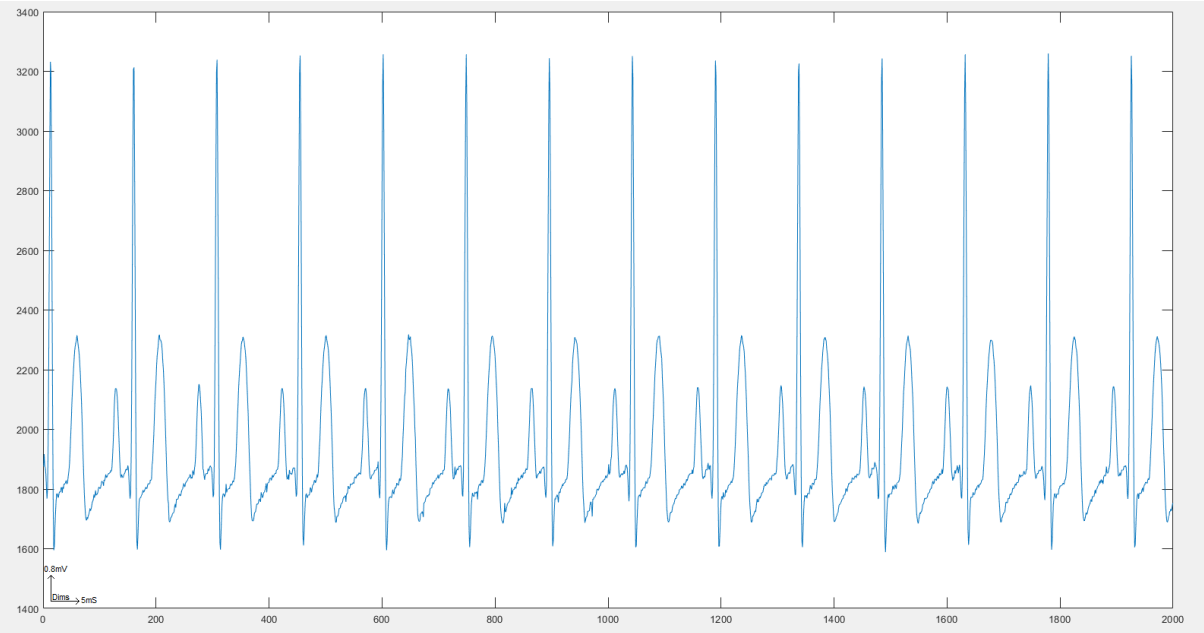


Figure 9. Normal 80 bpm data generated by the simulator and sampled by the ADC at 195 Hz for 10 seconds.

4-2. Preprocessing

This section describes preprocessing of the data acquired from the analog-to-digital converter (ADC). The purpose is to prepare the data for use by the artificial-intelligence model. The process includes normalization, peak detection, segmentation of heartbeats into windows, and adjustment of window lengths.

First, raw values from the ADC are processed. They must be normalized so that differences in scale are removed and the values lie on a common scale. Normalization uses the z-score method, in which the mean and standard deviation are calculated and the values standardized according to Equation 1.

$$z = (x - \mu) / \sigma$$

Equation 1. Normalization relationship.

In Equation 1, x is the data value, μ is the mean, and σ is the standard deviation. If the standard deviation is zero (all values are identical), only the mean is subtracted.

Next, R peaks representing heartbeats are identified. Appropriate thresholds and local-maximum detection algorithms are used to detect these peaks. Minimum peaks are also detected to locate other key points in the ECG. These points are used to identify local maxima within each window.

After peak detection, the signal is divided into windows, each corresponding to one heartbeat. These windows are determined using the identified R peaks and maxima. Each window contains the information associated with one heartbeat.

Finally, heartbeat-window lengths are adjusted so that they can be used by the artificial-intelligence model. Points are removed from or added to the windows as required. If a window exceeds the target length, excess points are removed from its beginning and end.

If a window is shorter than the target length, zero-valued points are added to its beginning and end. All heartbeat windows are ultimately combined into one output array suitable for input to artificial-intelligence models.

Thus, preprocessing transforms the raw data acquired by the microcontroller's ADC into a form appropriate for artificial-intelligence models. This makes preprocessing an essential stage in the analysis of vital signals.

As an example, Figure 10 shows an ECG signal containing a PVC, generated by a simulator and sampled by the microcontroller's ADC at 195 Hz over 10 seconds. Figure 11 presents the windows identified by the preprocessing algorithm.

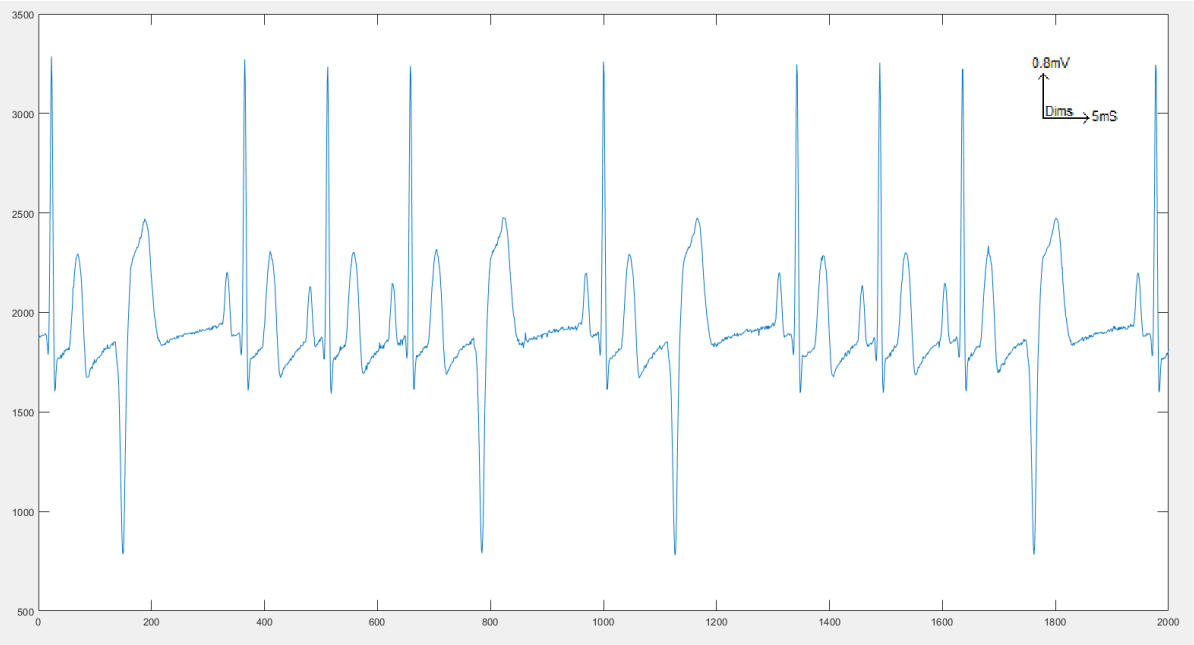


Figure 10. Simulated ECG signal containing a PVC, sampled by the ADC at 195 Hz over a 10-second interval.

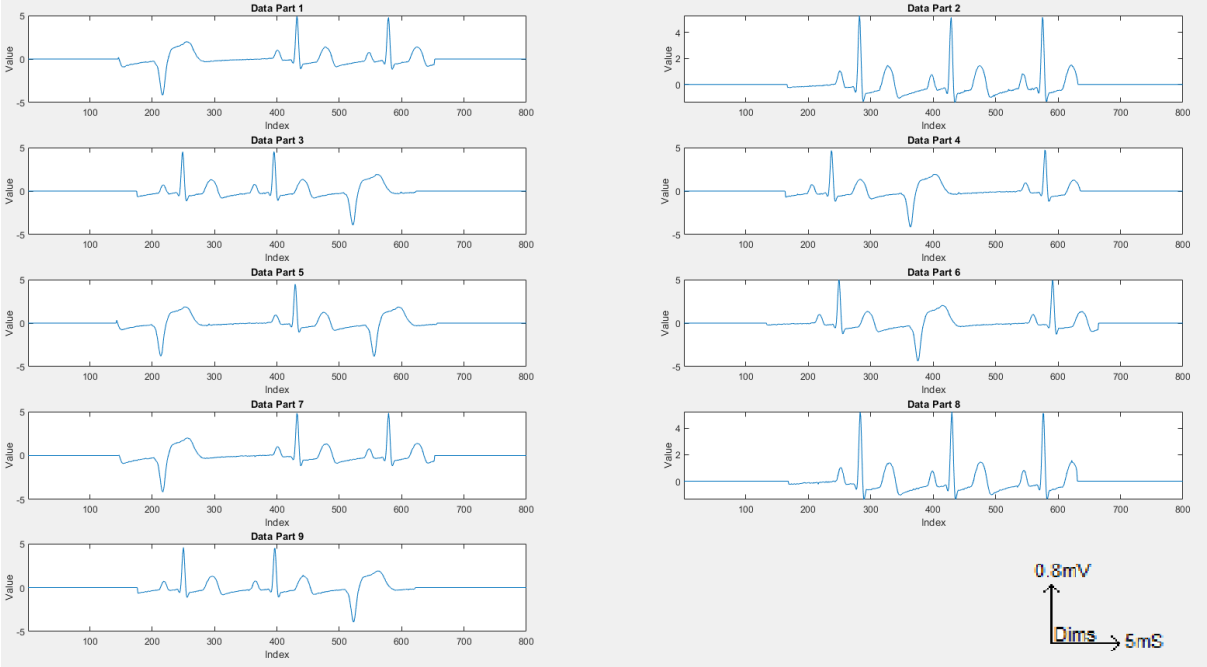


Figure 11. Windows identified by the preprocessing algorithm.

4-3. Artificial-Intelligence Model

This section describes in detail the preparation and training of the artificial-intelligence model for detecting different types of cardiac arrhythmias. The process includes the use of reference databases, training the model with TensorFlow, reducing model size through quantization, and deploying it on the ESP32-S3. The ultimate goal is to classify preprocessed ECG signals into categories including PVC, PAC, normal signal, and artifacts. The process is designed to combine high accuracy with the ability to run in a resource-constrained environment such as an ESP32-S3 microcontroller.

The MIT-BIH Arrhythmia Database is one of the best-known and most widely used sources for training models to detect cardiac arrhythmias. The database contains ECG signals recorded from 47 different patients and carefully annotated by cardiologists. Each recording contains detailed information about the arrhythmia type, its time of occurrence, and other characteristics.

Steps in Using the Database

Data loading and preprocessing: ECG signals are first read from the database and preprocessed through the following stages.

Normalization: Signals are normalized using the z-score method to reduce the effects of scale differences.

Noise removal: Unwanted interference, such as 50/60 Hz power-line noise and muscle-motion artifacts, is removed by digital filters, including low-pass and adaptive filters.

Heartbeat segmentation: The signal is divided into windows, each corresponding to one heartbeat and defined using the R peaks.

Data annotation: Recordings have been carefully annotated by specialists. The annotations identify the signal type, including PVC, PAC, normal beats, and artifacts. Examples include premature ventricular contraction (PVC), premature atrial contraction (PAC), normal cardiac beats without arrhythmia, and artifacts (unwanted noise present in the signal).

Splitting the data into training and test sets: The data are divided into two groups.

Training set: Approximately 80% of the data are used to train the model. This set includes different arrhythmia types and normal signals.

Test set: Approximately 20% of the data are used to evaluate model performance. They are not used in training and are reserved for final testing.

Data balancing: Because the sample counts for different arrhythmia classes are unequal (for example, PVC samples may be much less frequent than normal beats), techniques such as oversampling and undersampling are used to help the model learn all classes appropriately.

In addition to MIT-BIH, data from the AHA database are used. These consist of ECG signals recorded from real patients with locally prepared annotations. Including these data increases the diversity of the dataset and helps improve model accuracy.

The model is trained using the TensorFlow library, a powerful environment for developing deep-learning models.

Model Design and Training Steps

Selection of the model architecture: A convolutional neural network (CNN) is used to classify ECG signals because it can effectively extract features from time-series signals. Its architecture includes convolutional layers for extracting local features, pooling layers for reducing data dimensions and limiting overfitting, fully connected (FC) layers for final classification, and other components.

Model training: The model is trained using the training data. During this stage, the model weights are iteratively updated to reduce classification error. Optimization algorithms are also used to improve performance.

Model evaluation:

After training, the model is evaluated using the test set. Evaluation metrics include accuracy, sensitivity, specificity, and F1 score.

Model-Size Reduction (Quantization)

The model size must be reduced to deploy it on the ESP32. This is done through quantization. In this process, model weights and internal computations are converted from 32-bit floating-point precision to 8-bit integer precision. This can reduce model size to one-quarter of its original size and improve execution speed. Because the ESP32-S3 has limited RAM, this step is essential for deploying the model on the microcontroller.

Converting the Model to a Numerical Format

The final model is stored as a numerical array that can be executed directly on the ESP32-S3.

Evaluating Model Performance After Quantization

After reducing model size, the model is evaluated again to confirm that accuracy has been preserved to an acceptable degree.

Deploying the Model on the ESP32-S3

The reduced model is placed on the ESP32-S3. The microcontroller receives preprocessed data organized into windows of 800 points and supplies them to the model. The model then predicts whether each signal belongs to the PVC, PAC, normal, or artifact class.

This procedure enables the system to detect different arrhythmias and artifacts and to function as a tool for cardiac-health monitoring.

An example of the complete preprocessing and prediction process for an acquired signal is shown in Figures 15, 16, and 17. The original ADC signal, the preprocessed windows, and the model's predictions are presented in the following figures.

The original signal acquired by the ADC is shown in Figure 12; the preprocessed and segmented data are shown in Figure 13; and Figure 14 presents the predictions generated by the artificial-intelligence model for each window.

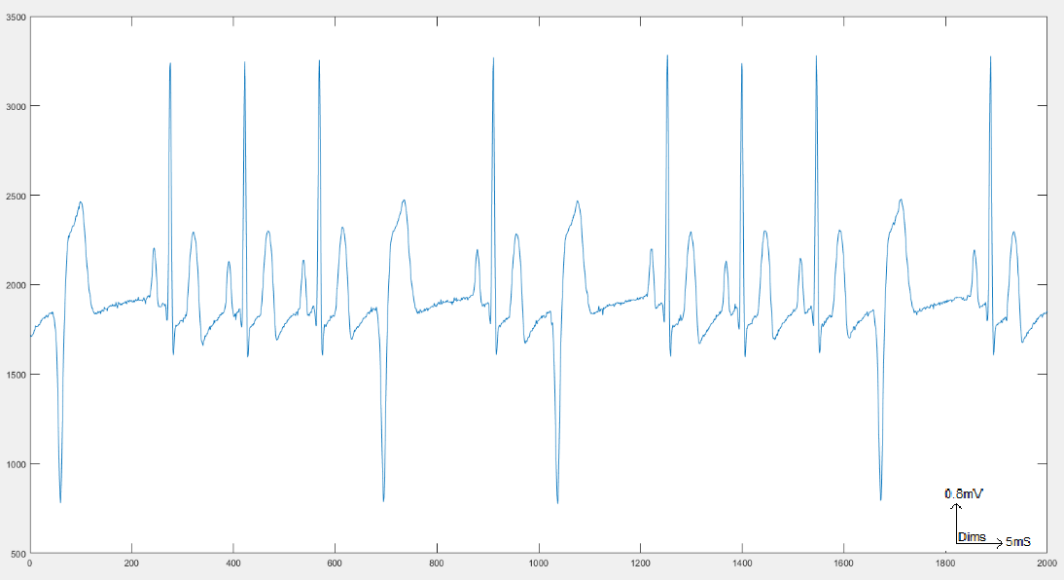


Figure 12. Simulator-generated ECG signal containing a PVC, acquired at 195 Hz over 10 seconds.

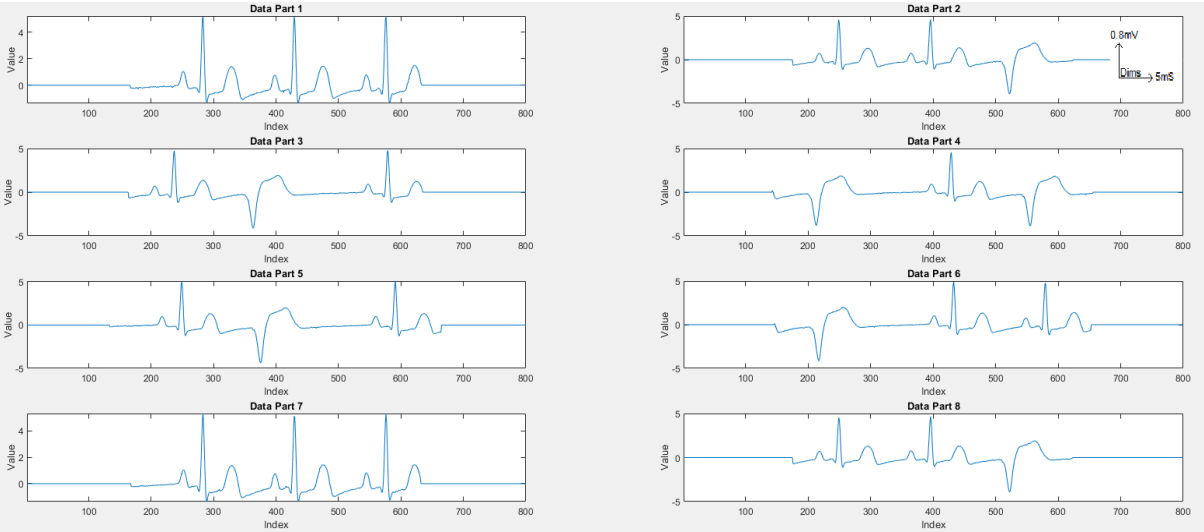


Figure 13. Windows detected by the preprocessing algorithm.

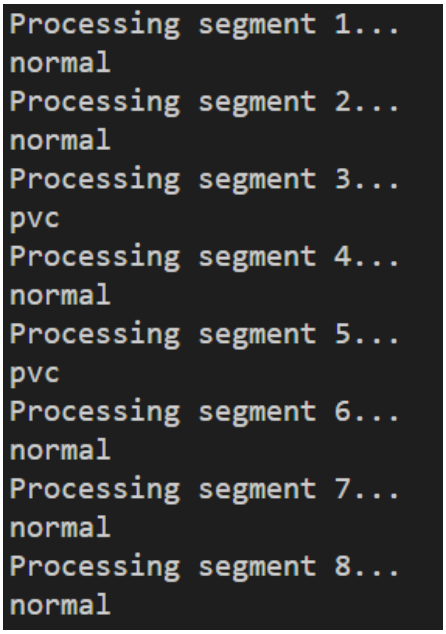


Figure 14. Artificial-intelligence model predictions for the individual windows detected by preprocessing.

4-4. Data Storage Upon Arrhythmia Detection

This section describes how preprocessed ECG data are identified and stored when an arrhythmia is detected. This stage is essential for preserving important events such as PVCs and PACs for later analysis. The main objective is to identify signals that indicate an arrhythmia automatically and save them in a file in the ESP32-S3 memory so that specialists can review them.

In this process, each window is sent to the artificial-intelligence model, which determines its arrhythmia class or the absence of an arrhythmia. The model generates a prediction for each window and identifies whether an arrhythmia is present. When an arrhythmia is detected, the corresponding information is flagged for storage.

After all windows have been classified, the system checks whether at least one window contains an arrhythmia. If an arrhythmia is found in even one window, the data for the entire 10-second period, including all its windows, are stored for later analysis. This strategy is intended to ensure that important events are not missed.

When an arrhythmia is detected, the relevant data are appended to a file. The file could be stored in the device's internal memory or external storage such as a memory card; in this design it is saved in the ESP32-S3 microcontroller's flash memory so that it can be read directly when needed. Append-mode storage adds new records without deleting previously saved information.

After data processing and storage, the system returns to a low-power state (Light Sleep) to reduce energy consumption. This is especially important for portable devices because it extends battery operating time. Only essential units, such as the RTC and ULP, remain active, and the system is ready to reactivate when required.

For example, Figure 15 shows an acquired ECG recording of approximately one minute. The arrhythmic intervals identified in this recording are shown in Figure 16.

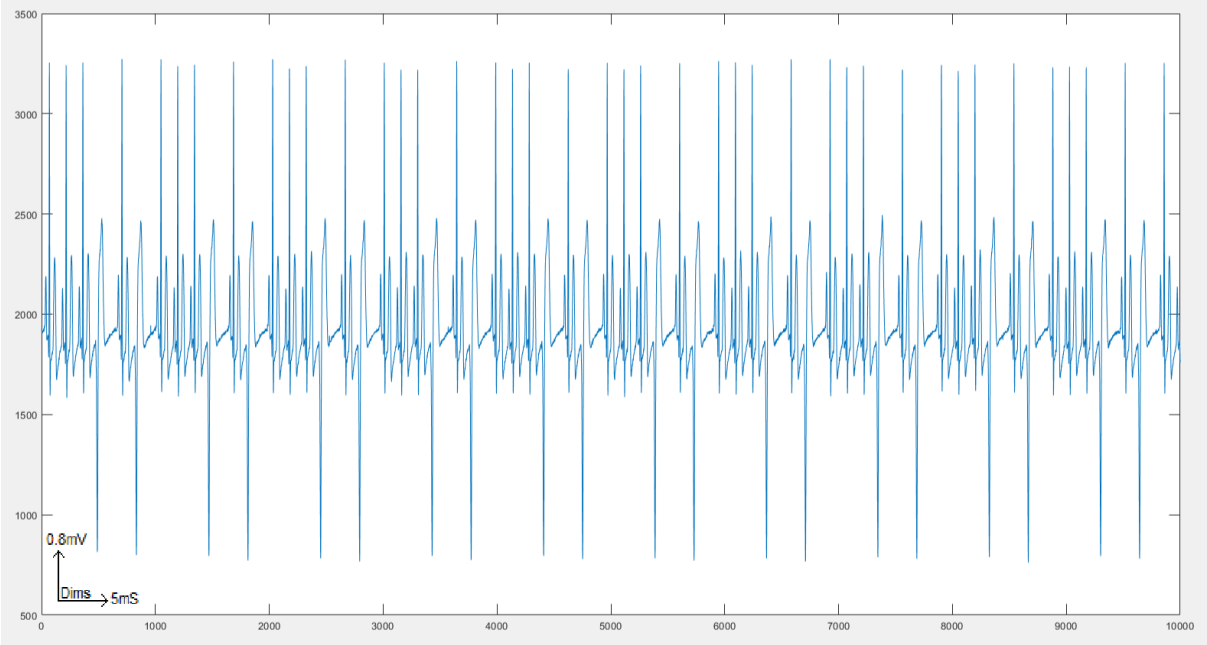


Figure 15. Simulator-generated ECG signal containing a PVC, acquired at 195 Hz over approximately 60 seconds.

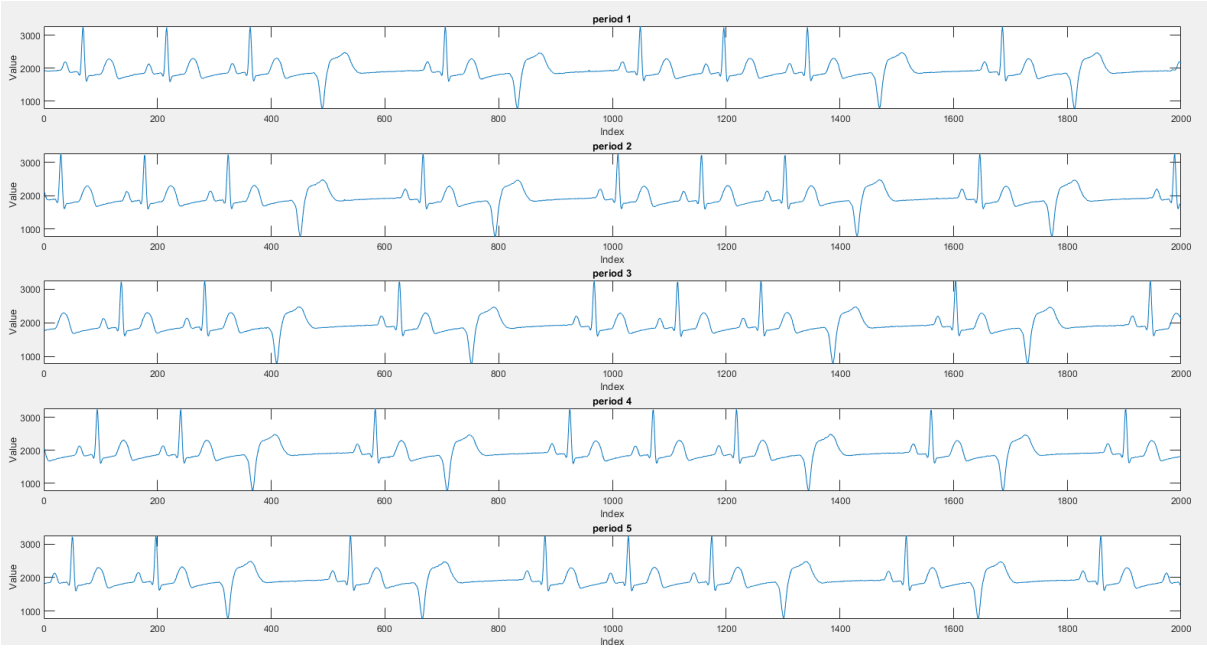


Figure 16. Detected and stored windows containing arrhythmia.

Chapter Five Summary and Conclusion

In this research, a wearable system was designed and implemented for continuous ECG monitoring based on an electronic front-end circuit and an artificial-intelligence algorithm, addressing the clinical need for timely detection of cardiac arrhythmias. The device can acquire the heart's electrical signal, amplify and digitize it, and analyze cardiac rhythm in real time in 10-second intervals using an optimized TensorFlow Lite model on a low-power microcontroller.

When arrhythmias such as PVC and PAC are detected, the relevant portion of the signal is stored in device memory together with the temporal information preceding and following the event. This allows the treating physician to make timely, accurate clinical decisions based on the patient's actual recorded data.

The principal advantages of the designed system include:

- Early detection of transient or irregular arrhythmias that are not identified by short-duration tests.
- Remote, long-term monitoring of patients without requiring attendance at healthcare facilities.
- Reduced diagnostic costs and improved quality of patient care.
- Portability, low power consumption, and ease of use in everyday life.

The reported system performance showed that careful hardware design combined with lightweight deep learning can provide a suitable basis for implementing arrhythmia-detection algorithms on embedded systems.

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